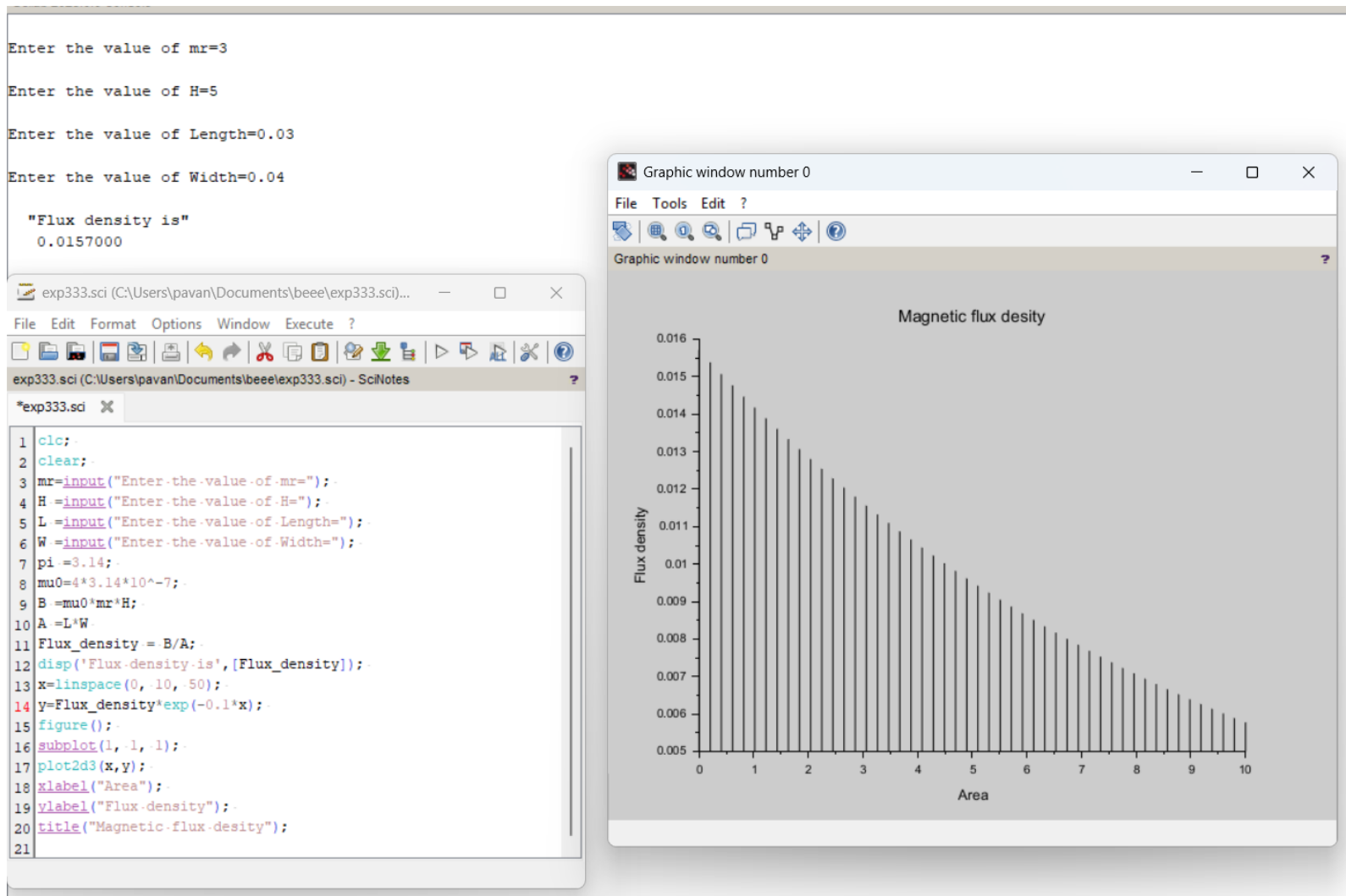


EXP NO:11

Determination and plotting of Magnetic Flux density in Ferromagnetic Materials with different (i) length, (ii) width and (iii) magnetic field intensity.



Calculation:

Exp: II

$$\mu_r = 3$$

$$H = 5$$

$$\text{Length} = 3 \text{ cm} = 0.03$$

$$\text{Width} = 4 \text{ cm} = 0.04$$

$$\text{Area: } A = L \cdot W$$

$$= 0.03 \times 0.04$$

$$= 0.0012 \text{ m}^2$$

$$\text{Magnetic flux density } B$$

$$\mu_0 = 4\pi \times 10^{-7} = 1.256 \times 10^{-6}$$

$$B = \mu_0 \cdot \mu_r \cdot H \Rightarrow (1.256 \times 10^{-6}) \times 3 \times 5 = 1.884 \times 10^{-5} \text{ T}$$

$$\text{Flux density} = \frac{B}{A} = \frac{1.884 \times 10^{-5}}{0.0012} = 1.57 \times 10^{-2} \text{ T/m}^2$$

$$\phi_0 = 0.0157 //$$

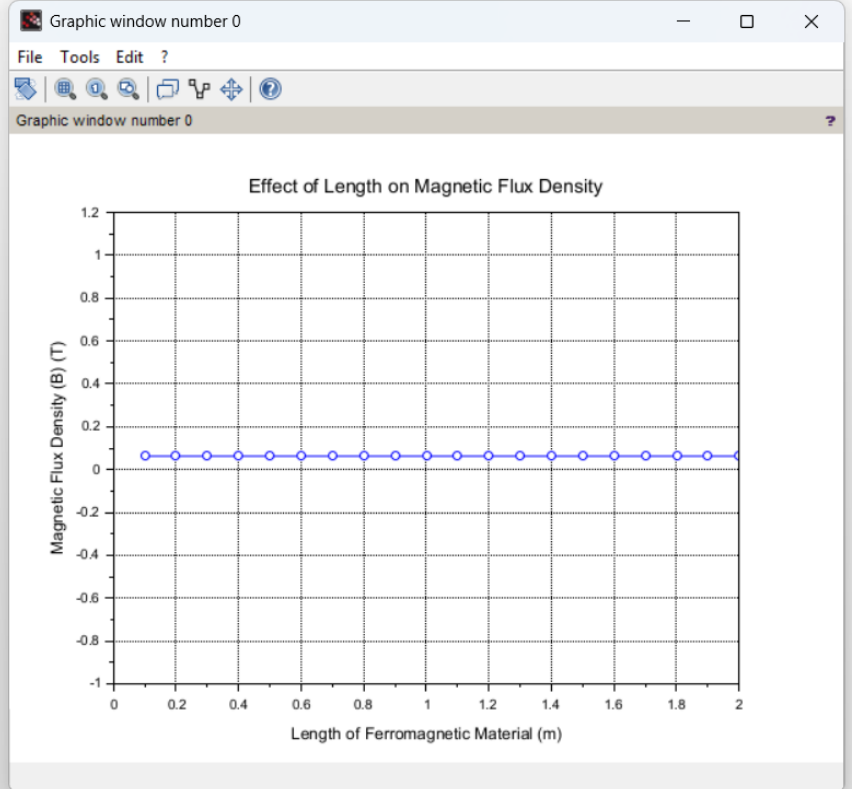
CASE:1

Examine the effect of length of a ferromagnetic material on the magnetic flux density (B).

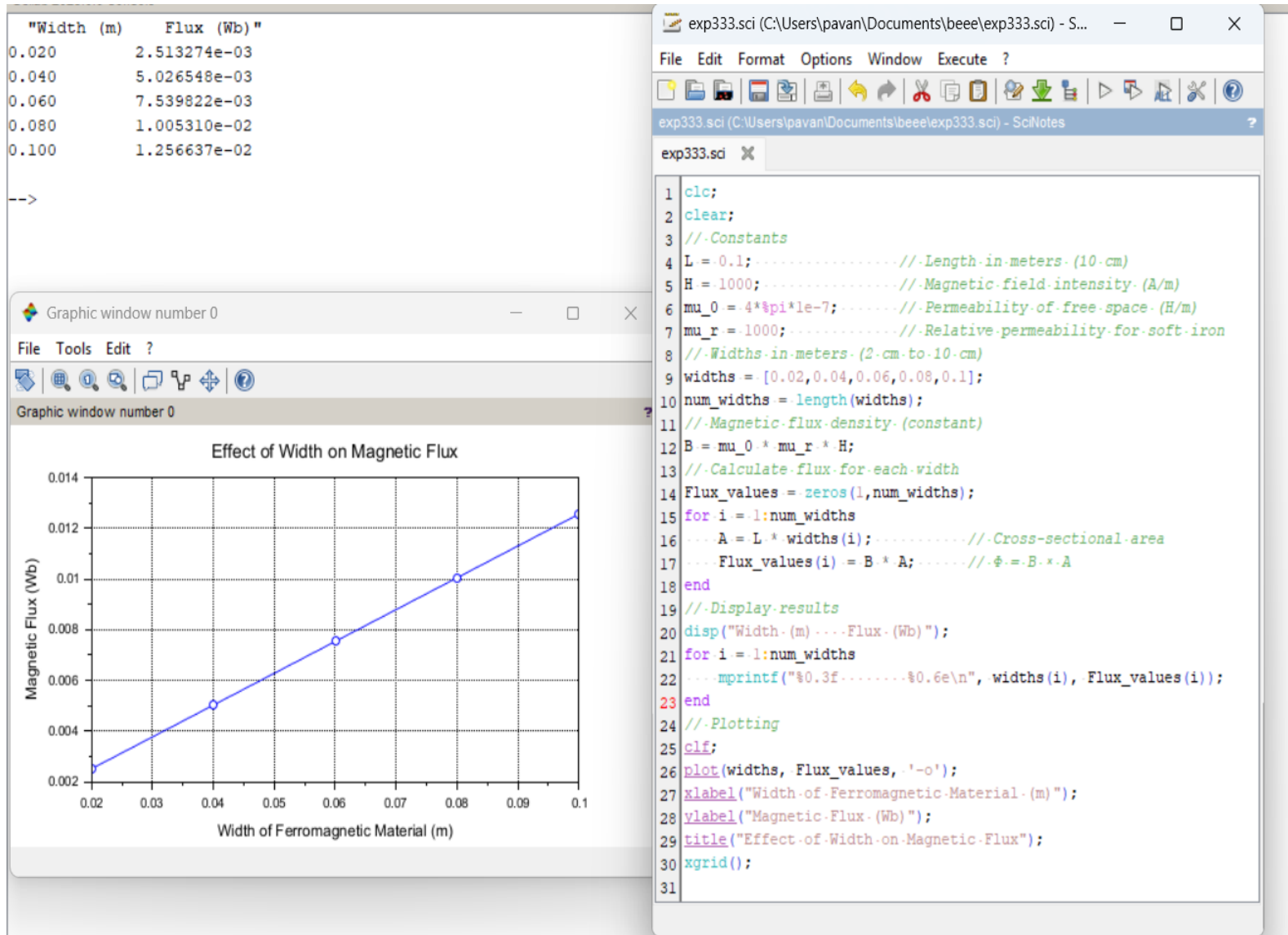
"Magnetic Flux Density (B): 0.0628319 Tesla"

-->

```
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci)...
File Edit Format Options Window Execute ?
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
exp333.sci
1 clc;
2 clear;
3
4 //Constants
5 mu_r = 500; //Relative permeability
6 H = 100; //Magnetic field intensity (A/m)
7 lengths = 0.1:0.1:2; //Lengths from 0.1 m to 2 m
8 mu_0 = 4*pi*1e-7; //Permeability of free space
9
10 //Magnetic flux density (independent of length)
11 B = mu_r * mu_0 * H;
12
13 //Display result
14 disp("Magnetic Flux Density (B): " + string(B) + " Tesla");
15
16 //Plotting (flat line)
17 clf;
18 plot(lengths, B*ones(length(lengths),1), '-o');
19 xlabel("Length of Ferromagnetic Material (m)");
20 ylabel("Magnetic Flux Density (B) (T)");
21 title("Effect of Length on Magnetic Flux Density");
22 xgrid();
23
```



CASE:2 To investigate how the magnetic flux density (B) in a ferromagnetic material changes with different widths at a constant length and magnetic field intensity.



case 3: To examine how the magnetic flux density (B) in a ferromagnetic material change with different values of magnetic field intensity at constant length and width.

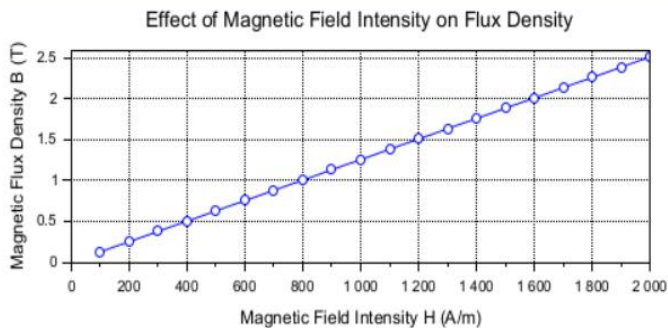
"Magnetic Field Intensity (A/m)	Flux Density (T)"
100	0.125664
200	0.251327
300	0.376991
400	0.502655
500	0.628319
600	0.753982
700	0.879646
800	1.005310
900	1.130973
1000	1.256637
1100	1.382301
1200	1.507964
1300	1.633628
1400	1.759292
1500	1.884956
1600	2.010619
1700	2.136283
1800	2.261947
1900	2.387610
2000	2.513274

Graphic window number 0

File Tools Edit ?



Graphic window number 0



```
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - S...
File Edit Format Options Window Execute ?
exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
exp333.sci X
1 clc;
2 clear;
3 L = 0.1; ..... // Length (m) -- constant
4 W = 0.05; ..... // Width (m) -- constant
5 mu_0 = 4*pi*1e-7; ..... // Permeability of free space (H/m)
6 mu_r = 1000; ..... // Relative permeability (soft iron)
7 // Range of magnetic field intensities (A/m)
8 H_values = 100:100:2000; // From 100 to 2000 A/m
9 num_H = length(H_values);
10 // Initialize array for B
11 B_values = zeros(1,num_H);
12 // Calculate B for each H
13 for i = 1:num_H
14 ... B_values(i) = mu_0 * mu_r * H_values(i);
15 end
16 // Display results
17 disp("Magnetic Field Intensity (A/m) .... Flux Density (T)");
18 for i = 1:num_H
19 ... mprintf("%d\t\t\t\t0.6f\n", H_values(i), B_values(i));
20 end
21 // Plotting
22 clf;
23 plot(H_values, B_values, '-o');
24 xlabel("Magnetic Field Intensity H (A/m)");
25 ylabel("Magnetic Flux Density B (T)");
26 title("Effect of Magnetic Field Intensity on Flux Density");
27 xgrid();
28
```

